

# ZHAOMIN WU

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## BIO

Dr. Zhaomin Wu is an NRF Postdoctoral Fellow and Research Fellow at the Department of Computer Science, National University of Singapore. He received his Ph.D. in Computer Science from NUS in 2024, advised by Prof. Bingsheng He. His research focuses on trustworthy AI, federated learning, privacy-preserving machine learning, LLM safety, multi-agent systems, and data management for collaborative learning.

He received the Singapore NRF Postdoctoral Award in 2026, a highly competitive national-level fellowship awarded by the National Research Foundation, Singapore, providing four-year salary support and an independent SGD 250,000 research grant. His work has been recognized with two ICLR 2026 Oral papers, each selected as top 1% of submissions, the SIGMOD 2023 Honorable Mention for Best Artifact, the NUS School of Computing Honorable Mention for Best Ph.D. Thesis Award, and the NUS Institute of Data Science Best Research Staff Award. As of May 2026, his work has received 2,710 citations on Google Scholar.

## EDUCATION

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>National University of Singapore (NUS), Singapore</b><br>Ph.D. in Computer Science<br>Thesis: Towards Practical Vertical Federated Learning<br><i>Honorable Mention for Best Ph.D. Thesis Award</i> | 07/2019 – 05/2024<br>Advisor: Prof. Bingsheng He |
| <b>Huazhong University of Science and Technology (HUST), China</b><br>B.Eng. in Computer Science and Technology, ACM class; GPA: 3.91/4.0, Rank: 10/281                                                | 09/2015 – 06/2019                                |

## EMPLOYMENT

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|---------------------------------------------------------------------------------------------|-------------------|
| <b>National University of Singapore (NUS), Singapore</b>                                    | 05/2024 – Present |
| • <i>NRF Postdoctoral Fellow</i> , Department of Computer Science                           | 04/2026 – Present |
| • <b>Principal Investigator (PI)</b> of an NRF-funded postdoctoral project                  |                   |
| • <i>Research Fellow</i> , Department of Computer Science<br>Project PI: Prof. Bingsheng He | 07/2025 – 03/2026 |
| • <i>Research Fellow</i> , Institute of Data Science<br>Project PI: Prof. Ng See Kiong      | 05/2024 – 06/2025 |

## SELECTED PUBLICATIONS

Zhaomin Wu, Haodong Zhao, Ziyang Wang, Jizhou Guo, Qian Wang, Bingsheng He. **LLM DNA: Tracing Model Evolution via Functional Representations.** (ICLR'26). **Oral presentation, top 1% of submissions.** Covered by [XinZhiYuan](#), [Sina Finance](#), [Sohu](#), [BAAI Hub](#), and [NICE Academic](#); NICE Academic video recorded 1,000+ views.

Zhaomin Wu, Mingzhe Du, See-Kiong Ng, Bingsheng He. **Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts.** (ICLR'26). **Oral presentation, top 1% of submissions.** Covered by [JiQiZhiXin](#) and [AITNT](#); AITNT post recorded 6,500+ clicks.

Zhaomin Wu, Junyi Hou, Bingsheng He. **VertiBench: Advancing Feature Distribution Diversity in Vertical Federated Learning Benchmarks.** (ICLR'24). *The first benchmark to systematically evaluate vertical federated learning under varying feature-distribution settings.*

Zhaomin Wu, Junhui Zhu, Qinbin Li, Bingsheng He. **DeltaBoost: Gradient Boosting Decision Trees with Efficient Machine Unlearning.** (SIGMOD'23). **Honorable Mention for Best Artifact, one of 3 selected from ~600 SIGMOD submissions.** *The first GBDT-like model to support efficient machine unlearning.*

Zhaomin Wu, Qinbin Li, Bingsheng He. **A Coupled Design of Exploiting Record Similarity for Practical Vertical Federated Learning.** (NeurIPS'22). *Enables practical vertical federated learning by exploiting inter-party record similarity instead of exact identifier matching.*

## HONORS AND AWARDS

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### Grants

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- **NRF Postdoctoral Award** 2026  
A highly competitive national-level fellowship awarded by the National Research Foundation (Singapore); four-year salary support and an independent SGD 250,000 research grant.
- **Google Cloud Research Credit Award** 2026  
Awarded by Google; USD 5,000 Google Cloud credits over one year.

### Research Awards and Professional Recognition

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- **SIGMOD Honorable Mention for Best Artifact** 2023  
Awarded by ACM SIGMOD; 3 selected among ~600 submissions.
- **Honorable Mention for Best Ph.D. Thesis Award** 2024  
Awarded by the School of Computing, National University of Singapore; 5 selected among ~150 Ph.D. graduates.
- **Best Research Staff Award** 2025  
Awarded by the Institute of Data Science, National University of Singapore; 2 selected among ~50 research staff.
- **Dean's Graduate Research Excellence Award** 2023  
Awarded by the School of Computing, National University of Singapore; 10 awardees among ~150 graduate researchers.
- **ICML 2026 Gold Reviewer Award** 2026  
Awarded by the International Conference on Machine Learning (ICML); top 25% among reviewers.

### Undergraduate Honors

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- **Outstanding Students** 2016  
Awarded by Huazhong University of Science and Technology; top 1%.
- **National Scholarship** 2017  
Awarded by the Ministry of Education of China; top 1% at HUST.
- **Outstanding Undergraduate Thesis** 2019  
Awarded by Huazhong University of Science and Technology; top 5%.
- **Merit Student** 2016  
Awarded by Huazhong University of Science and Technology; top 5%.
- **Meritorious Winner, Mathematical Contest in Modeling** 2018  
Awarded by COMAP; top 10% worldwide.

## RESEARCH IMPACT platform metrics recorded in May 2026

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### Open-Source Systems and Benchmarks

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- [FedTree](#) (MLSys'23): an open-source federated learning system for tree-based models; it supports reproducible and privacy-preserving GBDT training across parties. 152 GitHub stars and 41 forks.
- [VertiBench](#) (ICLR'24): a public benchmark for systematically evaluating vertical federated learning under varying feature-diversity settings; it standardizes evaluation beyond single synthetic data splits.
- [WikiDBGGraph](#) (ICDE'26): a large-scale database graph of Wikidata for collaborative learning; it releases 100,000 database nodes and 17 million edges for database-silo learning research.
- [LLM DNA / RepTrace](#) (ICLR'26 Oral): an open-source toolkit for extracting functional DNA signatures and tracing LLM evolution, with a public project website and PyPI package.
- [OARF](#) (ACM TIST'22): a benchmark suite for characterizing federated learning systems; it provides reusable workloads for system-level FL comparison.

## Publicity and Public Outreach

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### *LLM DNA: Tracing Model Evolution via Functional Representations* (ICLR'26 Oral).

**Impact:** makes hidden LLM lineage, fine-tuning, and distillation relationships traceable from model behavior rather than model weights.

- Media article by [XinZhiYuan \(新智元\)](#), a major Chinese AI media outlet: “Testing DNA for Large Models: Hidden Lineage from Fine-tuning and Distillation Can No Longer Hide” (给大模型验DNA, 微调、蒸馏的隐藏血缘藏不住了).
- Repost by [Sina Finance \(新浪财经\)](#): a major Chinese finance and business-news portal operated by Sina.
- Repost by [Sohu \(搜狐\)](#): a major Chinese web portal and technology-news distribution platform.
- Repost by [BAAI Hub \(智源社区\)](#): an academic AI community platform by BAAI.
- Public outreach hosted by [NICE Academic \(NICE学术\)](#) on Bilibili: “Doing Paternity Testing for Large Models: LLM DNA and Model Evolution Graph” (给大模型做亲子鉴定: LLM 的DNA 与模型演化图谱).
- The NICE Academic video recorded 1,000+ views.

### *Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts* (ICLR'26 Oral).

**Impact:** reframes LLM honesty evaluation by showing that deception-like inconsistency can appear under ordinary, benign prompts.

- Media article by [JiQiZhiXin \(机器之心\)](#), a major Chinese AI and frontier-technology media platform: “ICLR 2026 Oral: Large Models Can Still ‘Lie’ Without Prompt Induction” (ICLR 2026 Oral | 没人诱导, 大模型也会骗人).
- Repost by [AITNT](#): a Chinese AI news portal; the republication post recorded 6,500+ clicks.

### *Mining Intrinsic Rewards from LLM Hidden States for Efficient Best-of-N Sampling* (KDD'26).

**Impact:** provides a lightweight reward-modeling path by using LLM hidden states, reducing reward-model cost for Best-of-N inference.

- Media article by [XinZhiYuan \(新智元\)](#): “Reward Models Are Changing: 0.005% Parameters, Faster Inference, Stronger Performance” (奖励模型变天! 0.005%参数量推理速度翻倍, 性能还更强).
- Repost by [Sina Finance \(新浪财经\)](#): a major Chinese finance and business-news portal operated by Sina.
- Repost by [Sohu \(搜狐\)](#): a major Chinese web portal and technology-news distribution platform.
- Repost by [BAAI Hub \(智源社区\)](#): an academic AI community platform by BAAI.
- Repost by [Tencent Cloud Developer Community \(腾讯云开发者社区\)](#): a developer-facing technical content platform by Tencent Cloud.

## INVITED TALKS & KEYNOTES

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### 2026

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**Invited Talk:** “When Data and Models Stay Hidden: Toward Trustworthy AI Collaboration” May 2026  
Université de Montréal (UdeM) and Mila; hosted by Prof. Bang Liu

**Keynote:** “Bridging Data Silos with Practical Federated Learning” Jan. 2026  
AAAI 2026 FLCA Workshop

### 2025

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**Invited Talk:** “Towards Practical Vertical Federated Learning Systems” May 2025  
DASFAA 2025 Trust Day; on behalf of Prof. Bingsheng He

**Invited Talk:** “Bridging Private Data Silo with Machine Learning” Feb. 2025  
NUS Open House 2025

## TEACHING & MENTORSHIP

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### Teaching Assistant

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|                                            |             |
|--------------------------------------------|-------------|
| Introduction to Operating Systems (CS2106) | Fall 2022   |
| Software Testing (CS4218)                  | Spring 2022 |
| Neural Networks and Deep Learning (CS5242) | Fall 2021   |
| Programming Methodology (II) (CS2030)      | Spring 2021 |
| Big Data Systems for Data Science (CS4225) | Fall 2020   |
| Data Structures and Algorithms (CS2040S)   | Spring 2020 |
| Programming Methodology (CS1010E)          | Fall 2019   |

### Mentored Students

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|-------------------------------------------------------------------------------------------------------------------|-------------|
| Jiayi Li, master student, NUS                                                                                     | 2025 – 2026 |
| Ziyang Qiu, master student, NUS                                                                                   | 2025 – 2026 |
| Lifan Hu, undergraduate student, NUS                                                                              | 2025 – 2026 |
| Qian Yu, undergraduate student, NUS                                                                               | 2025 – 2026 |
| Ziyang Wang, master student, NUS                                                                                  | 2024 – 2025 |
| Haodong Zhao, visiting Ph.D. student, Shanghai Jiao Tong University                                               | 2024 – 2025 |
| Zhen Qin, visiting Ph.D. student, Zhejiang University<br><i>Current: Assistant Professor, Zhejiang University</i> | 2024 – 2025 |
| Jizhou Guo, undergraduate student, Shanghai Jiao Tong University                                                  | 2024        |
| Junyi Hou, master student, NUS<br><i>Current: Ph.D. student, NUS</i>                                              | 2023 – 2024 |
| Junhui Zhu, master student, NUS                                                                                   | 2022        |

## FULL PUBLICATION LIST

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\*Equal contribution, <sup>†</sup>Corresponding author

2026

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[ICLR'26] Zhaomin Wu, Haodong Zhao, Ziyang Wang, Jizhou Guo, Qian Wang, Bingsheng He. LLM DNA: Tracing Model Evolution via Functional Representations. In The Fourteenth International Conference on Learning Representations (ICLR), 2026. (*Oral, 1%*)

[ICLR'26] Zhaomin Wu, Mingzhe Du, See-Kiong Ng, Bingsheng He. Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts. In The Fourteenth International Conference on Learning Representations (ICLR), 2026. (*Oral, 1%*)

[ICDE'26] Zhaomin Wu<sup>\*</sup>, Ziyang Wang<sup>\*</sup>, Bingsheng He. WikiDBGraph: Large-Scale Database Graph of Wikidata for Collaborative Learning. In Proceedings of the IEEE International Conference on Data Engineering (ICDE), 2026.

[KDD'26] Jizhou Guo, Zhaomin Wu<sup>†</sup>, Hanchen Yang, Philip S. Yu. Mining Intrinsic Rewards from LLM Hidden States for Efficient Best-of-N Sampling. In Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026.

[arXiv'26] Haodong Zhao, Jinming Hu, Zhaomin Wu<sup>†</sup>, Zongru Wu, Wei Du, Junyi Hou, Caibei Zhao, Zhuosheng Zhang, Bingsheng He, Gongshen Liu<sup>†</sup>. ProtegoFed: Backdoor-Free Federated Instruction Tuning with Interspersed Poisoned Data. arXiv preprint arXiv:2603.00516, 2026.

[WWW'26] Yicheng Zhang, Zhen Qin, Zhaomin Wu, Shuiguang Deng<sup>†</sup>. Personalized Federated Fine-Tuning for LLMs via Data-Driven Heterogeneous Model Architectures. In Proceedings of the ACM Web Conference (WWW), 2026. (*Oral, 9%*)

2025

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[EMNLP'25] Zhaomin Wu<sup>\*</sup>, Jizhou Guo<sup>\*</sup>, Junyi Hou, Bingsheng He, Lixin Fan, Qiang Yang. Model-based Large Language Model Customization as Service. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[arXiv'25] Zhaomin Wu, Zhen Qin, Junyi Hou, Haodong Zhao, Qinbin Li, Bingsheng He, Lixin Fan. Vertical Federated Learning in Practice: The Good, the Bad, and the Ugly. arXiv preprint arXiv:2502.08160, 2025.

[ACL'25] Zhen Qin, Zhaomin Wu<sup>†</sup>, Bingsheng He, Shuiguang Deng<sup>†</sup>. Federated Data-Efficient Instruction Tuning for Large Language Models. In Findings of the Association for Computational Linguistics: ACL 2025.

[arXiv'25] Haodong Zhao, Jidong Li, Zhaomin Wu<sup>†</sup>, Tianjie Ju, Zhuosheng Zhang, Bingsheng He, Gongshen Liu<sup>†</sup>. Disagreements in Reasoning: How a Model's Thinking Process Dictates Persuasion in Multi-Agent Systems. arXiv preprint arXiv:2509.21054, 2025.

2024

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[NeurIPS'24] Zhaomin Wu, Junyi Hou, Yiqun Diao, Bingsheng He. Federated Transformer: Multi-Party Vertical Federated Learning on Practical Fuzzily Linked Data. In Advances in Neural Information Processing Systems (NeurIPS), 2024.

[ICLR'24] Zhaomin Wu, Junyi Hou, Bingsheng He. VertiBench: Advancing Feature Distribution Diversity in Vertical Federated Learning Benchmarks. In The Twelfth International Conference on Learning Representations (ICLR), 2024.

2023

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[SIGMOD'23] Zhaomin Wu, Junhui Zhu, Qinbin Li, Bingsheng He. DeltaBoost: Gradient Boosting Decision Trees with Efficient Machine Unlearning. In Proceedings of the ACM on Management of Data (SIGMOD), 2023. *(Best Artifact Honorable Mention, 3 out of ~600)*

[MLSys'23] Qinbin Li, Zhaomin Wu, Yanzheng Cai, Yuxuan Han, Ching Man Yung, Tianyuan Fu, Bingsheng He. FedTree: A Federated Learning System for Trees. In Proceedings of Machine Learning and Systems (MLSys), 2023.

2022

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[NeurIPS'22] Zhaomin Wu, Qinbin Li, Bingsheng He. A Coupled Design of Exploiting Record Similarity for Practical Vertical Federated Learning. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

[TBD'22] Zhaomin Wu, Qinbin Li, Bingsheng He. Practical Vertical Federated Learning with Unsupervised Representation Learning. IEEE Transactions on Big Data, 2022.

[TIST'22] Sixu Hu, Yuan Li, Xu Liu, Qinbin Li, Zhaomin Wu, Bingsheng He. The OARF Benchmark Suite: Characterization and Implications for Federated Learning Systems. ACM Transactions on Intelligent Systems and Technology (TIST), 2022.

2021

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[TKDE'21] Qinbin Li, Zeyi Wen, Zhaomin Wu, Sixu Hu, Naibo Wang, Yuan Li, Xu Liu, Bingsheng He. A Survey on Federated Learning Systems: Vision, Hype and Reality for Data Privacy and Protection. IEEE Transactions on Knowledge and Data Engineering (TKDE), 2021.

2020

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[AAAI'20] Qinbin Li, Zhaomin Wu, Zeyi Wen, Bingsheng He. Privacy-Preserving Gradient Boosting Decision Trees. In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2020.

## PROFESSIONAL SERVICES

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2026

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**Conferences:** ICML (Reviewer), NeurIPS (Reviewer)

**Journals:** ACM Computing Survey (Reviewer), TPDS (Reviewer), TKDE (Reviewer), TMC (Reviewer)

2025

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**Conferences:** ICLR (Reviewer), ICML (Reviewer), NeurIPS (Reviewer), AAAI (Reviewer), EMNLP (Reviewer)

**Journals:** TDSC (Reviewer), TPDS (Reviewer), TKDE (Reviewer), TAI (Reviewer)

2024

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**Conferences:** ICLR (Reviewer), NeurIPS (Reviewer), KDD (Reviewer), AISTATS (Reviewer)

**Journals:** TKDD (Reviewer), JPDC (Reviewer), TNNLS (Reviewer)

2023

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**Conferences:** NeurIPS (Reviewer), SIGMOD (Artifact Reviewer)

**Journals:** TPAMI (Reviewer), TKDE (Reviewer), TNNLS (Reviewer), IJCV (Reviewer), IoT-J (Reviewer)

2022

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**Conferences:** PAKDD (Reviewer)

**Journals:** TPDS (Reviewer), TIST (Reviewer)

2021

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**Conferences:** PAKDD (Reviewer)

2020

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**Tutorial:** IJCAI (Tutorial Speaker)

## LANGUAGE

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**Chinese (Mandarin)**

Native speaker

**English**

Professional working proficiency